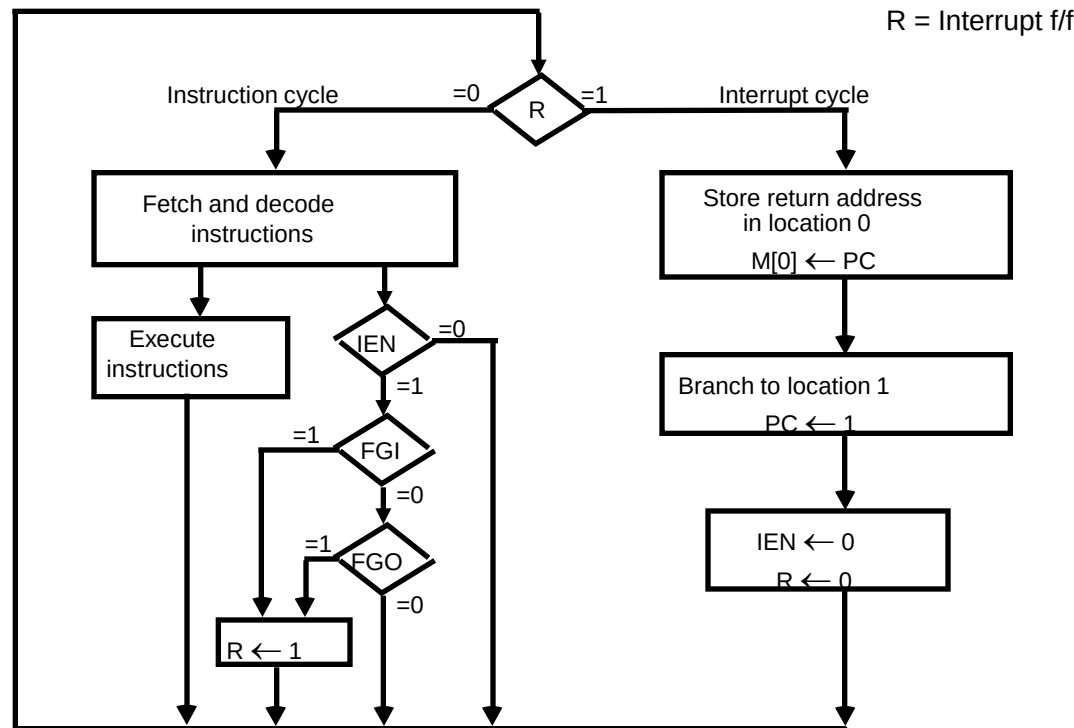


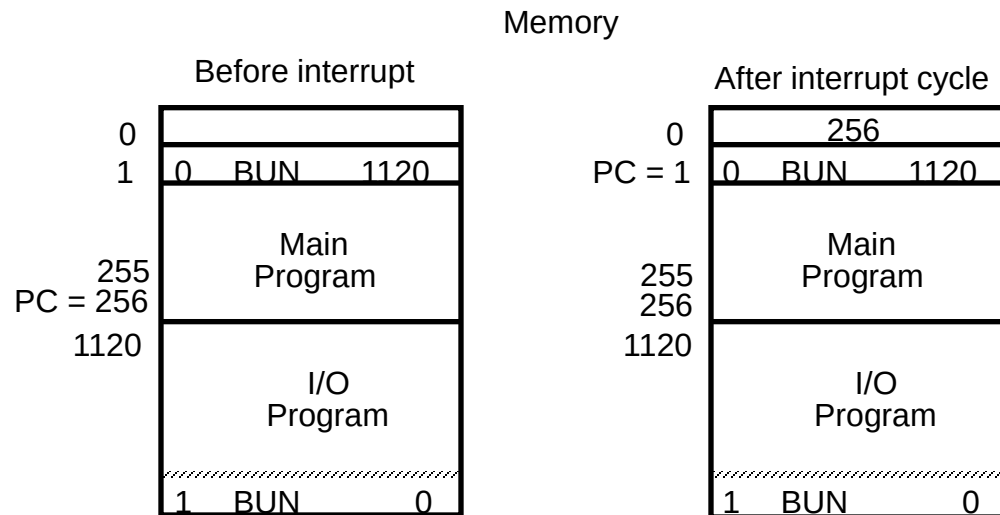
FLOWCHART FOR INTERRUPT CYCLE



- The interrupt cycle is a HW implementation of a branch and save return address operation.
- At the beginning of the next instruction cycle, the instruction that is read from memory is in address 1.
- At memory address 1, the programmer must store a branch instruction that sends the control to an interrupt service routine.
- The instruction that returns the control to the original program is "indirect BUN 0"



REGISTER TRANSFER OPERATIONS IN INTERRUPT CYCLE



Register Transfer Statements for Interrupt Cycle

- $R \leftarrow F/F \leftarrow 1$ if $IEN (FGI + FGO)T_0'T_1'T_2' \Leftrightarrow T_0'T_1'T_2' (IEN)(FGI + FGO): R \leftarrow 1$
- The fetch and decode phases of the instruction cycle must be modified $\rightarrow$ Replace T_0, T_1, T_2 with $R'T_0, R'T_1, R'T_2$
- The interrupt cycle :
 - $RT_0: AR \leftarrow 0, TR \leftarrow PC$
 - $RT_1: M[AR] \leftarrow TR, PC \leftarrow 0$
 - $RT_2: PC \leftarrow PC + 1, IEN \leftarrow 0, R \leftarrow 0, SC \leftarrow 0$



COMPLETE COMPUTER DESCRIPTION

Flowchart of Operations

